

Part 1: Perimeter and Area Relationships of a Rectangle

Optimization: the process of finding values that make a given quantity the greatest (or least) possible given certain conditions.

Investigation 1:

In this investigation you will determine how to find the maximum area of a rectangle with a fixed perimeter.

Remember: $Perimeter\ of\ a\ Rectangle = 2(l + w)$ **or** $2l + 2w$

$Area\ of\ a\ Rectangle = length \times width$

Instructions: Brandon needs to find the dimensions that will maximize the rectangular area of an enclosure with a perimeter of 32 meters.

a) Copy and complete the following table to determine the dimensions of the rectangle that maximizes the area of a rectangle that has a perimeter of 32 meters.

Rectangle	Width	Length	Perimeter	Area
1	1	15	32	15
2	2	14		
3				
4				
5				
6				
7				
8				

b) What are the dimensions of the rectangle with the maximum area? _____

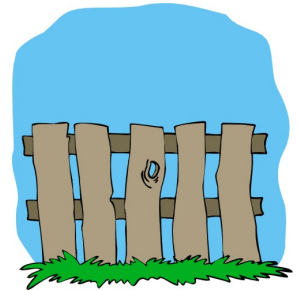
c) What is the maximum area? _____

d) Describe the shape of the rectangle:

Problem 1

Your neighbour has asked for your advice about building his garden. He wants to fence the largest rectangular garden possible with 12 metres of fencing. What are the dimensions of the garden with the maximum area?

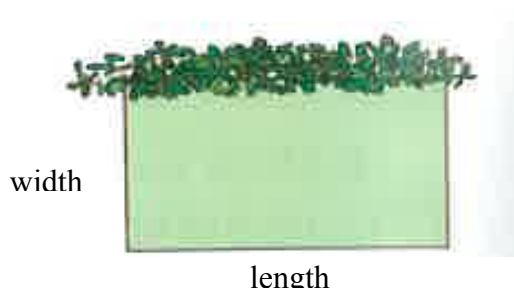
How can you organize the possible shapes so you don't miss one that might give the maximum area?



Answer:

Investigation 2: In this investigation you will determine how to find the maximum area of a rectangle with a fixed sum of the lengths of three sides.

Instructions: Brandon's customer decides to use an existing hedge as one of the boundaries for the enclosure. This means that he will only use the prefabricated fencing on three sides of the rectangular enclosure. The client still wants the enclosure to have the greatest area possible. Brandon has 32 meters of prefabricated fencing.



a) Do you think Brandon will be able to enclose more, less, or the same amount of area now that the hedge is being used on one side?

b) What shape do you think will have the maximum area?

c) Copy and complete the chart to determine the dimensions of the rectangle that has the maximum area.

Rectangle	Width	Length	Sum of lengths of three sides	Area
1	1	30	32	30
2	2	28	32	
3				
4				
5				
6				
7				
8				
9				
10				

d) What are the dimensions of the rectangle with maximum/optimal area? _____

e) What is the maximum area? _____

f) Describe the shape of the rectangle: _____

Problem 2

A farmer is adding a rectangular corral to the side of a barn. The barn will form one side of the rectangle. The farmer has 16 meters of fencing to use. Conduct an investigation to determine the dimensions of the corral with maximum area.

The dimensions are:

Consolidation:

The dimensions of a rectangle that has maximum area depend on the number of sides to be fenced.

When fencing all four sides of a rectangle, the optimal shape is a _____.

When only three sides are being fenced, the relationship between the

length and the width of the rectangle is $l =$ _____.

Apply What You Learned

1) What dimensions will provide the maximum area for a rectangle with each perimeter?

a) 20 m

c) 50 m

b) 36 m

d) 83 m

2. Cam is building a fenced-in play yard for his small children against one wall of his house. He has 12 m of fencing. What whole-number dimensions produce the play yard with the greatest area?